Remark

(1) An angle between two non-zero vectors $\vec{a}$ and $\vec{b}$ is found by the following formula $\theta = \cos^{-1}\left(\frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}\right)$.

(2) $\vec{a}$ and $\vec{b}$ are said to be parallel if the angle between them is 0 or π.

(3) $\vec{a}$ and $\vec{b}$ are said to be perpendicular if the angle between them is $\frac{\pi}{2}$ or $\frac{3\pi}{2}$.

Property

(1) Let $\vec{a}$ and $\vec{b}$ be any two nonzero vectors. Then

- $\vec{a}\cdot\vec{b} = 0$ if and only if $\vec{a}$ and $\vec{b}$ are perpendicular to each other.
- $\vec{a}\times\vec{b} = \vec{0}$ if and only if $\vec{a}$ and $\vec{b}$ are parallel to each other.

(2) If $\vec{a},\vec{b}$, and $\vec{c}$ are any three vectors and α is a scalar, then

$$\vec{a}\cdot\vec{b} = \vec{b}\cdot\vec{a},\ (\vec{a}+\vec{b})\cdot\vec{c} = \vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c},\ (\alpha\vec{a})\cdot\vec{b} = \alpha(\vec{a}\cdot\vec{b}) = \vec{a}\cdot(\alpha\vec{b});$$

$$\vec{a}\times\vec{b} = -(\vec{b}\times\vec{a}),\ (\vec{a}+\vec{b})\times\vec{c} = \vec{a}\times\vec{c}+\vec{b}\times\vec{c},\ (\alpha\vec{a})\times\vec{b} = \alpha(\vec{a}\times\vec{b}) = \vec{a}\times(\alpha\vec{b}).$$

6.3.2 Application of dot and cross products in plane Trigonometry

We apply the concepts of dot and cross products of two vectors to derive a few formulae in plane trigonometry.

Example 6.1 (Cosine formulae)

With usual notations, in any triangle *ABC*, prove the following by vector method.

(i) $a^2 = b^2 + c^2 - 2bc\cos A$ (ii) $b^2 = c^2 + a^2 - 2ca\cos B$

(iii) $c^2 = a^2 + b^2 - 2ab\cos C$

Solution

With usual notations in triangle *ABC*, we have $\overrightarrow{BC} = \vec{a}, \overrightarrow{CA} = \vec{b}$ and $\overrightarrow{AB} = \vec{c}$. Then $|\overrightarrow{BC}| = a$, $|\overrightarrow{CA}| = b$, $|\overrightarrow{AB}| = c$ and $\overrightarrow{BC}+\overrightarrow{CA}+\overrightarrow{AB} = \vec{0}$.

So, $\overrightarrow{BC} = -\overrightarrow{CA} - \overrightarrow{AB}$.

Then applying dot product, we get

$$\overrightarrow{BC}\cdot\overrightarrow{BC} = (-\overrightarrow{CA}-\overrightarrow{AB})\cdot(-\overrightarrow{CA}-\overrightarrow{AB})$$

$$\Rightarrow\ |\overrightarrow{BC}|^2 = |\overrightarrow{CA}|^2 + |\overrightarrow{AB}|^2 + 2\overrightarrow{CA}\cdot\overrightarrow{AB}$$

$$\Rightarrow\ a^2 = b^2 + c^2 + 2bc\cos(\pi - A)$$

$$\Rightarrow\ a^2 = b^2 + c^2 - 2bc\cos A.$$

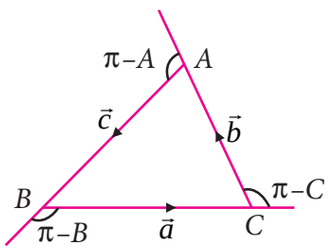


Fig. 6.6

The results in (ii) and (iii) are proved in a similar way.

Example 6.2

With usual notations, in any triangle *ABC*, prove the following by vector method.

(i) $a = b\cos C + c\cos B$ (ii) $b = c\cos A + a\cos C$

(iii) $c = a\cos B + b\cos A$

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